PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Most Likely - Main Test - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Choose 3 of the following 4 problems. Complete every part of each chosen problem. These are the four strongest practice matches to the announced topics, not confirmed exam questions.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 5 - Question 23.7

23.7 •• Two protons, starting several meters apart, are aimed directly at each other with speeds of 2.00×10^5 m/s, measured relative to the earth. Find the maximum electric force that these protons will exert on each other.

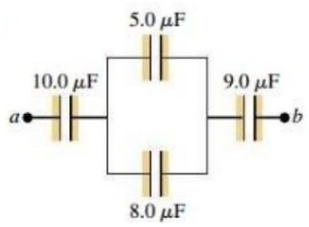
2. [____] Homework 4 - Question 22.19

22.19 •• A hollow, conducting sphere with an outer radius of 0.250 m and an inner radius of 0.200 m has a uniform surface charge density of $+6.37 \times 10^{-6}$ C/m^2. A charge of $-0.500\ \mu$C is now introduced at the center of the cavity inside the sphere. (a) What is the new charge density on the outside of the sphere? (b) Calculate the strength of the electric field just outside the sphere. (c) What is the electric flux through a spherical surface just inside the inner surface of the sphere?

3. [____] Homework 7 - Question 24.14

24.14 • **Figure E24.14** shows a system of four capacitors, where the potential difference across *ab* is 50.0 V. (a) Find the equivalent capacitance of this system between *a* and *b*. (b) How much charge is stored by this combination of capacitors? (c) How much charge is stored in each of the 10.0-μF and the 9.0-μF capacitors?

Figure **E24.14**



4. [____] Tutorial 6 - Question 47

47. A parallel-plate capacitor in air has a plate separation of 1.50 cm and a plate area of 25.0 cm^2. The plates are charged to a potential difference of 250 V and disconnected from the source. The capacitor is then immersed in distilled water. Determine (a) the charge on the plates before and after immersion, (b) the capacitance and potential difference after immersion, and (c) the change in energy of the capacitor. Assume the liquid is an insulator.

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Most Likely - Ranked Follow-up - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions in the ranked study order. These continue the shortlist after the main test; this longer paper is extra practice, not the announced exam format.

Write all rough calculations and final answers on separate paper. Show your working.

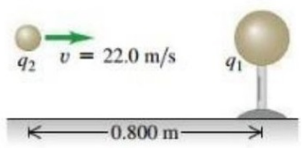
Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 5 - Question 23.5

23.5 •• A small metal sphere, carrying a net charge of $q_1 = -2.80\ \mu C$, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -7.80\ \mu C$ and mass 1.50 g, is projected toward q_1. When the two spheres are 0.800 m apart, q_2, is moving toward q_1 with speed 22.0 m/s (**Fig. E23.5**). Assume that the two spheres can be treated as point charges. You can ignore the force of gravity. (a) What is the speed of q_2 when the spheres are 0.400 m apart? (b) How close does q_2 get to q_1?

Figure **E23.5**

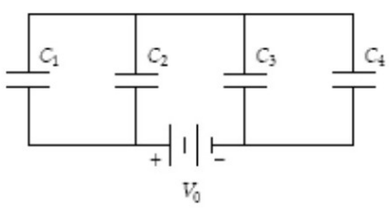


2. [____] Tutorial 6 - Question 69

69. A parallel-plate capacitor of plate separation d is charged to a potential difference ΔV_0. A dielectric slab of thickness d and dielectric constant κ is introduced between the plates while the battery remains connected to the plates. (a) Show that the ratio of energy stored after the dielectric is introduced to the energy stored in the empty capacitor is $U/U_0 = \kappa$. Give a physical explanation for this increase in stored energy. (b) What happens to the charge on the capacitor? (Note that this situation is not the same as in Example 26.7, in which the battery was removed from the circuit before the dielectric was introduced.)

3. [____] Tutorial 7 - Question 1

1. What is the total energy stored by C_3 when $C_1 = 50\ \mu F$, $C_2 = 30\ \mu F$, $C_3 = 36\ \mu F$, $C_4 = 12\ \mu F$, and $V_0 = 30$ V?



a. 6.3 mJ
b. 25 mJ
c. 57 mJ
d. 1.6 mJ
e. 14 mJ

4. [____] Tutorial 4 - Question 7

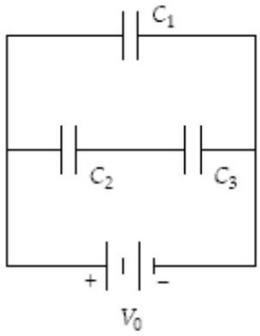
7. A 4.0-pC point charge is placed at the center of a hollow (inner radius = 2.0 cm, outer radius = 4.0 cm) conducting sphere which has a net charge of 4.0 pC. Determine the magnitude of the electric field at a point which is 6.0 cm from the point charge.
   a. 35 N/C
   b. 25 N/C
   c. 30 N/C
   d. 20 N/C
   e. 10 N/C

5. [____] Tutorial 4 - Question 9

9. A point charge of 6.0 nC is placed at the center of a hollow spherical conductor (inner radius = 1.0 cm, outer radius = 2.0 cm) which has a net charge of –4.0 nC. Determine the resulting charge density on the inner surface of the conducting sphere.
   a. +4.8 μC/m^2
   b. –4.8 μC/m^2
   c. –9.5 μC/m^2
   d. +9.5 μC/m^2
   e. –8.0 μC/m^2

6. [____] Tutorial 7 - Question 2

2. What is the potential difference across C_2 when $C_1 = 5.0\ \mu$F, $C_2 = 15\ \mu$F, $C_3 = 30\ \mu$F, and $V_0 = 24$ V?



   a. 21 V
   b. 19 V
   c. 16 V
   d. 24 V
   e. 8.0 V

7. [____] Tutorial 7 - Question 7

7. A parallel plate capacitor of capacitance C_0 has plates of area A with separation d between them. When it is connected to a battery of voltage V_0, it has charge of magnitude Q_0 on its plates. It is then disconnected from the battery and the space between the plates is filled with a material of dielectric constant 3. After the dielectric is added, the magnitudes of the charge on the plates and the potential difference between them are
   a. $\frac{1}{3}Q_0, \frac{1}{3}V_0$.
   b. $Q_0, \frac{1}{3}V_0$.
   c. Q_0, V_0.
   d. $Q_0, 3V_0$.
   e. $3Q_0, 3V_0$.

8. [____] Homework 7 - Question 24.36

24.36 • A budding electronics hobbyist wants to make a simple 1.0-nF capacitor for tuning her crystal radio, using two sheets of aluminum foil as plates, with a few sheets of paper between them as a dielectric. The paper has a dielectric constant of 3.0, and the thickness of one sheet of it is 0.20 mm. (a) If the sheets of paper measure 22 × 28 cm and she cuts the aluminum foil to the same dimensions, how many sheets of paper should she use between her plates to get the proper capacitance? (b) Suppose for convenience she wants to use a single sheet of posterboard, with the same dielectric constant but a thickness of 12.0 mm, instead of the paper. What area of aluminum foil will she need for her plates to get her 1.0 nF of capacitance? (c) Suppose she goes high-tech and finds a sheet of Teflon of the same thickness as the posterboard to use as a dielectric. Will she need a larger or smaller area of Teflon than of posterboard? Explain.

PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Most Likely - Bonus Practice - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Bonus practice only. Attempt all questions in ranked study order. The actual announcement specifies two bonus problems; these seven questions cover the shortlisted alternatives.

Write all rough calculations and final answers on separate paper. Show your working.

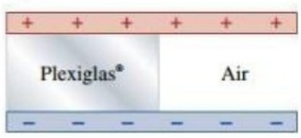
Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Homework 7 - Question 24.66

24.66 •• A parallel-plate capacitor is made from two plates 12.0 cm on each side and 4.50 mm apart. Half of the space between these plates contains only air, but the other half is filled with Plexiglas® of dielectric constant 3.40 (**Fig. P24.66**). An 18.0-V battery is connected across the plates. (a) What is the capacitance of this combination? (*Hint:* Can you think of this capacitor as equivalent to two capacitors in parallel?) (b) How much energy is stored in the capacitor? (c) If we remove the Plexiglas® but change nothing else, how much energy will be stored in the capacitor?

Figure **P24.66**



2. [____] Homework 4 - Question 22.21

22.21 •• The electric field at a distance of 0.145 m from the surface of a solid insulating sphere with radius 0.355 m is 1750 N/C. (a) Assuming the sphere's charge is uniformly distributed, what is the charge density inside it? (b) Calculate the electric field inside the sphere at a distance of 0.200 m from the center.

3. [____] Textbook 15e - Question 22.47

22.47 •• **CALC** A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

$$\rho(r) = \rho_0\left(1 - \frac{r}{R}\right) \qquad \text{for } r \le R$$

$$\rho(r) = 0 \qquad \text{for } r \ge R$$

where $\rho_0 = 3Q/\pi R^3$ is a positive constant. (a) Show that the total charge contained in the charge distribution is Q. (b) Show that the electric field in the region $r \ge R$ is identical to that produced by a point charge Q at $r = 0$. (c) Obtain an expression for the electric field in the region $r \le R$. (d) Graph the electric-field magnitude E as a function of r. (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

4. [____] Textbook 15e - Question 22.50

22.50 •• CALC A solid insulating sphere has radius R and carries positive charge distributed throughout its volume. The charge distribution has spherical symmetry but varies with radial distance r from the center of the sphere. The volume charge density is $\rho(r) = \rho_0(1 - r/R)$, where ρ_0 is a constant with units of C/m^3. (a) Derive an expression for the electric field as a function of r for $r < R$. (b) Repeat part (a) for $r > R$. (c) At what value of r, in terms of R, does the electric field have its maximum value?

5. [____] Textbook 15e - Question 24.34

24.34 •• Two identical air-filled parallel-plate capacitors C_1 and C_2, each with capacitance C, are connected in series to a battery that has voltage V. While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of one of the capacitors, completely filling the space between them. Let U_0 be the total energy stored in the two capacitors without the dielectric and U be the total energy stored after the dielectric is inserted. In terms of K, what is the ratio U/U_0? Does the total stored energy increase, decrease, or stay the same after the dielectric is inserted?

6. [____] Textbook 15e - Question 24.35

24.35 •• Two identical air-filled parallel-plate capacitors C_1 and C_2 are connected in series to a battery that has voltage V. The charge on each capacitor is Q_0. While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of capacitor C_1, completely filling the space between them. In terms of K and Q_0, what is the charge on capacitor C_1 after the dielectric is inserted? Does the charge on C_1 increase, decrease, or stay the same?

7. [____] Textbook 15e - Question 24.36

24.36 •• Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in series to a battery that has voltage V; $C_1 = 3.00\ \mu F$ and $C_2 = 6.00\ \mu F$. The electric field between the plates of capacitor C_2 is E_{02}. While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K = 4$ is inserted between the plates of capacitor C_1, completely filling the space between them. After the dielectric is inserted in C_1, the electric field between the plates of capacitor C_2 is E_2. (a) What is the ratio E_2/E_{02}? When the dielectric is inserted into C_1, does the electric field in C_2 increase, decrease, or remain the same? (b) Repeat the calculation in part (a) for the two capacitors connected to the battery in parallel.